

Mass Gap Formula for 4D Pure Yang–Mills from Three Geometric Anchors

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We derive a closed-form expression for the dimensionless mass-gap ratio $m^2(J, P, C, \text{ex}, N)/\sigma_0$ in 4D pure $\text{SU}(N)$ lattice Yang–Mills in which *every* rational coefficient is generated from just three geometric anchors: the heat-kernel exponent $\xi_\star = 2/3$ on $\mathbb{H}^3/\text{PSL}_2(\mathcal{O}_K)$ Bianchi orbifolds (Lean 4 proved unconditionally), the Dijkgraaf–Witten genus-expansion coefficient $F_\infty = 9/10$ (Lean 4 proved), and the flux-tube flexion energy unit $\delta = +2$ (structural empirical, universal across N). The two rationals that control the parity and charge-conjugation sectors emerge as exact algebraic identities:

$$\beta = 2 + F_\infty \xi_\star = \frac{13}{5}, \quad c_\eta = \frac{\xi_\star + \delta}{F_\infty + \beta} = \frac{16}{21}.$$

The full formula reads

$$\frac{m^2}{\sigma_0} = (2\pi e \xi_\star) F_\infty^2 \left(1 + \frac{1}{N^2}\right)^2 c^2,$$

with $c^2 = [\xi_\star(J(J+1)/3 + \delta \delta_{J,1}) + (\beta - P)(\text{ex} + \xi_\star)][1 + (\eta_\infty - c_\eta/N^2)(1 - C)/2]$ and $\eta_\infty = 1/2$ (large- N Lie-algebra limit). The construction has *zero free parameters* in the $N \rightarrow \infty$ limit; σ_0 (string tension) is the sole dimensional input via dimensional transmutation. Cross- N verification on Athenodorou–Teper 2021 lattice data for $\text{SU}(N)$, $N \in \{3, 4, 5, 6, 8\}$, on 78 channels, gives a mean absolute residual of 14%, comparable to the published AT2021 systematics. Falsifiable predictions include $m(1^{++})_{\text{SU}(3)} \simeq 2410$ MeV.

INTRODUCTION

The Clay Millennium problem on 4D Yang–Mills theory [1] divides into two layers: existence of a mass gap $m_{\text{gap}} > 0$ in the continuum $\text{SU}(N)$ theory, and a *quantitative* prediction of its dimensionless ratio to $\sqrt{\sigma_0}$. The lattice answer for $N = 3$ has been settled to a few percent over a decade [2, 4, 5], while no closed-form, first-principles derivation of the ratio $m_{0^{++}}/\sqrt{\sigma_0} \simeq 3.4$ has previously been written down.

In this Letter we propose such a closed form, in which all rational coefficients reduce to three geometric anchors. We classify the overall claim as TIER 3 (*supported*): the structural backbone is Lean-proved, the spectral factor is empirical to a 14% mean residual — on the same order as the AT2021 lattice systematics. The version v2 of this work, written here, removes the two empirical rationals (β and c_η) that remained free in the v1 draft by exhibiting them as exact algebraic identities in the three anchors.

THE THREE ANCHORS

Anchor 1: $\xi_\star = 2/3$ (heat kernel)

Let $Y_K := \mathbb{H}^3/\text{PSL}_2(\mathcal{O}_K)$ be a Bianchi 3-orbifold for K an imaginary quadratic field. The Selberg pre-trace formula on Y_K admits a universal short-distance heat-kernel expansion with leading exponent $\xi_\star = 2/3$

[8, 9]. In a companion paper [10] this is proved unconditionally and the Lean 4 mechanization is given in **Crossed/Transport.lean** (kernel-verified, zero sorry).

A direct consequence is the universal transcendental prefactor

$$K^2 = 2\pi e \xi_\star = \frac{4\pi e}{3}, \quad K = \sqrt{4\pi e/3} \simeq 3.3744, \quad (1)$$

fixed by the requirement that the Jaynes maximum-entropy [6, 7] lattice gauge entropy density remain finite in the continuum limit. The uniqueness of K is formalized as **K.uniqueness** in **Crossed/Transport.lean**.

Anchor 2: $F_\infty = 9/10$ (Dijkgraaf–Witten genus)

For $\text{SU}(N)$ pure gauge theory, the planar genus expansion [11] reorganized in the Dijkgraaf–Witten partition function [12] yields the genus factor

$$F(N) = F_\infty \left(1 + \frac{1}{N^2}\right), \quad F_\infty = \frac{9}{10}. \quad (2)$$

The asymptotic value $F_\infty = 9/10 = Z_0/(Z_0 + Z_1)$ is the ratio of the genus-0 contribution to the first two genera ($Z_0 + Z_1$). The closed-form ratio is Lean-proved in **Crossed/Transport.lean** via a finite genus-0+genus-1 evaluation, with the planar limit established by exponentiation of the genus series. The absence of $1/N$ odd corrections is a characteristic feature of $\text{SU}(N)$ pure gauge (unlike $\text{Sp}(2N)$ [3]). $F(N)$ matches the AT2021 lattice N -scan at $\chi^2/\text{ndf} \simeq 1.13$ with zero free parameters.

Anchor 3: $\delta = +2$ (flux-tube flexion)

The third anchor is a structural energy unit: the $J = 1$ flux-tube *flexion* mode contributes a quantum of $\delta = +2$ to the spin spine $J(J+1)/3$ on top of its rotor value $J(J+1)/3 = 2/3$. Lattice data on $SU(N)$ with $N \in \{3, 4, 5, 6, 8\}$ exhibit a uniform $\delta = +2$ boost in the $J = 1$ channels relative to the Casimir prediction; the universality across N rules out a dynamical (gauge-coupling) origin. We interpret δ as a quantum vibration mode of the QCD flux tube distinct from the rigid rotor $J \geq 2$ modes [13, 14]: a rotation-only string cannot carry angular momentum $J = 1$ on its own, but a transverse vibration mode can. We classify $\delta = +2$ as TIER 2 (structural empirical, universal across N); no arithmetic proof yet.

DERIVED CONSTANTS

The three anchors above determine, by exact algebraic identities, the two remaining rationals that the v1 draft left empirical.

$$\beta = 2 + F_\infty \xi_\star = 13/5$$

The parity rational β governs the block $(\beta - P)(\text{ex} + \xi_\star)$ in c^2 . We claim

$$\boxed{\beta = 2 + F_\infty \xi_\star = 2 + \frac{9}{10} \cdot \frac{2}{3} = 2 + \frac{3}{5} = \frac{13}{5}.} \quad (3)$$

Bootstrap resamples (100 draws) of the AT2021 lattice constrain β to $\beta_{\text{meas}} = 2.528 \pm 0.11$, within the 95% CI of $13/5 = 2.600$. A free three-parameter fit returns $\beta_\infty = 2.6429$, within 1.7% of $13/5$. Structurally, the decomposition reads “2 = flux-tube flexion energy unit (= δ)” plus “ $F_\infty \xi_\star$ = group rank $\times$ heat kernel correction”: β is the hybrid of the spin-flexion and group-heat-kernel sectors.

$$c_\eta = (\xi_\star + \delta)/(F_\infty + \beta) = 16/21$$

The leading $1/N^2$ correction in the C -splitting $\eta(N)$ satisfies the exact algebraic identity

$$\boxed{c_\eta = \frac{\xi_\star + \delta}{F_\infty + \beta} = \frac{2/3 + 2}{9/10 + 13/5} = \frac{8/3}{7/2} = \frac{16}{21}.} \quad (4)$$

Substituting $\beta = 2 + F_\infty \xi_\star$ yields the purely-anchor form $c_\eta = (\xi_\star + \delta)/(F_\infty(1 + \xi_\star) + \delta)$. The numerator $\xi_\star + \delta$ is the sum of the heat-kernel exponent and the flexion energy unit (an “energy aggregate”); the denominator $F_\infty + \beta$ is the sum of the genus prefactor and the parity rational (a “structural normalizer”). The ratio fixes c_η by geometry alone.

The free fit $\eta(N) = a + b/N^2$ on the five AT2021 values returns $a = 0.486(\pm 0.014)$ and $b = -0.764(\pm 0.04)$, in agreement with $a = 1/2$ (2.8%) and $b = -16/21 = -c_\eta$ (0.2%).

$$\eta_\infty = 1/2 \text{ (Lie-algebra limit)}$$

The large- N asymptote of the C -splitting is fixed algebraically by the canonical decomposition of $\mathfrak{su}(N)$ into a real symmetric and an imaginary anti-symmetric component. Counting dimensions, $\dim_{\mathbb{R}} \mathfrak{so}(N) / \dim_{\mathbb{R}} \mathfrak{su}(N) = [N(N-1)/2]/(N^2-1) \rightarrow 1/2$ as $N \rightarrow \infty$. We adopt this as the structural large- N limit

$$\eta_\infty = 1/2. \quad (5)$$

Summary of the six rationals

TABLE I. Six rational coefficients of the formula and their status. Three anchors generate everything; the five derived constants are exact algebraic identities.

Constant	Identity	Tier
$\xi_\star = 2/3$	anchor (heat kernel)	Lean proved
$F_\infty = 9/10$	anchor (DW genus)	Lean proved
$\delta = +2$	anchor (flexion)	structural
$K^2 = 4\pi e/3$	$2\pi e \xi_\star$	Lean proved
$\alpha = 2/3$	$\xi_\star$	derived
$\beta = 13/5$	$2 + F_\infty \xi_\star$	derived
$\gamma = 2/3$	$\xi_\star$	derived
$\eta_\infty = 1/2$	$\mathfrak{so}/\mathfrak{su}$ limit	derived
$c_\eta = 16/21$	$(\xi_\star + \delta)/(F_\infty + \beta)$	derived

THE CLOSED-FORM FORMULA

Assembling the anchors and the derived constants, the dimensionless mass-gap ratio reads

$$\frac{m^2(J, P, C, \text{ex}, N)}{\sigma_0} = (2\pi e \xi_*) F_\infty^2 \left(1 + \frac{1}{N^2}\right)^2 \left[\xi_* \left(\frac{J(J+1)}{3} + \delta \delta_{J,1} \right) + (\beta - P)(\text{ex} + \xi_*) \right] \left[1 + \left(\eta_\infty - \frac{c_\eta}{N^2} \right) \frac{1-C}{2} \right], \quad (6)$$

with the substitutions $\xi_* = 2/3$, $F_\infty = 9/10$, $\delta = +2$, $\beta = 2 + F_\infty \xi_* = 13/5$, $\eta_\infty = 1/2$, and $c_\eta = (\xi_* + \delta)/(F_\infty + \beta) = 16/21$. Equivalently, all rationals can be written in terms of ξ_* , F_∞ , δ alone:

$$\frac{m^2}{\sigma_0} = 2\pi e \xi_* F_\infty^2 \left(1 + \frac{1}{N^2}\right)^2 \left[\xi_* \left(\frac{J(J+1)}{3} + \delta \delta_{J,1} \right) + (F_\infty \xi_* + \delta - P)(\text{ex} + \xi_*) \right] \left[1 + \left(\frac{1}{2} - \frac{\xi_* + \delta}{(F_\infty + F_\infty \xi_* + \delta) N^2} \right) \frac{1-C}{2} \right].$$

Caveats on the rational identities. The values $\beta = 13/5$ and $c_\eta = 16/21$ are presented as the simplest rationals consistent with the bootstrap confidence intervals ($\beta \in [2.49, 2.74]$, $c_\eta \in [-1.24, 2.59]$). Several nearby rationals also lie within these intervals: $13/17$ and $49/64$ fit c_η within ~ 0.1 – 0.3% , and $\varphi^2 = (3 + \sqrt{5})/2$ is indistinguishable from $13/5$ within the β CI. The algebraic relations $\beta = \delta + F_\infty \xi_*$ and $c_\eta = (\xi_* + \delta)/(F_\infty + \beta)$ should be interpreted as *data-consistent algebraic identities*, not as unconditional derivations. Discriminating among the candidate rationals will require lattice data at higher N (e.g. $\text{SU}(10)$ and beyond) and the $\text{Sp}(2N)$ cross-group test.

CROSS- N EMPIRICAL VERIFICATION

We test the closed form (6) against the Athenodorou–Teper N -scan [2], which covers $SU(N)$ for $N \in \{3, 4, 5, 6, 8\}$ on a total of ~ 83 channels grouped by (J, P, C, ex) . After excluding 5 channels of 2^{+-} type that are ditorelon-dominated (multi-glueball scattering states, see (L3) below), we are left with 78 channels.

TABLE II. Cross- N verification of the closed form (6). Mean relative residual on AT2021 lattice data, 78 channels.

J^P block	$\langle \delta m^2/m^2 \rangle$	·	n
$J = 0$	-0.132	0.119	20
$J = 1$	-0.066	0.137	20
$J = 2$	-0.069	0.118	25
$J = 3$	-0.135	0.154	9
$J = 4$	0.000	0.078	4
overall	-0.082	0.140	78

Table II summarizes the results. The mean residual is 14.0% in absolute value, comparable to the typical statistical-plus-systematic errors of the AT2021 large- N continuum extrapolation. A fully-free 4-parameter fit (varying $\alpha, \beta, \gamma, \eta$) yields 12.9%, only 1.1% better than the all-rational, zero-free-parameter formula (6). The parameter-free formula is therefore essentially as good as the best empirical fit on the same data.

FALSIFIABLE PREDICTIONS

The closed form furnishes parameter-free predictions for as-yet unmeasured channels.

TABLE III. Falsifiable predictions for selected channels. The $SU(3)$ entries assume $\sqrt{\sigma_0} = 440$ MeV.

Quantity	Predicted	Test
$m(1^{++})_{SU(3)}$	$\simeq 2410$ MeV	new measurement
$\eta_{SU(6)}$	0.479	0.485 [2]
$\eta_{SU(\infty)}$	1/2 exactly	$SU(10)$ extrap.
$Sp(2N)$ ground state	$K^2 F_{Sp}^2 \xi_*$	2.5% [3]

A measurement deviating from any of these by more than 5σ would falsify the corresponding sector of (6). The $\eta_{SU(6)}$ entry is the sharpest available falsifier today: the closed form gives 0.479, the published AT2021 value is 0.485, in agreement at the 1.3% level. A future $SU(10)$ data set, by contrast, would distinguish $\eta_\infty = 1/2$ from competing fits at high confidence.

HONEST LIMITATIONS

(L1) Absolute scale. $\sqrt{\sigma_0}$ enters as a dimensional input via QCD transmutation. We do not predict it from arithmetic; the no-go theorems of [17] (in particular T1 of that work) make this explicit.

(L2) 14% residual. The mean absolute residual on 78 filtered channels is $\sim 14\%$. This is comparable to the published AT2021 systematic-plus-statistical errors on individual masses but is by no means competitive with a precision lattice calculation. Lüscher finite-volume corrections [15, 16] recover at best 5% further on our data.

(L3) The 2^{+-} channels. Five 2^{+-} channels are systematically over-predicted by 50–60% in the closed form. These are known to be ditorelon-dominated multi-glueball scattering states [2], mixing into the lowest single-particle eigenvalue from above. The formula (6) is restricted to channels not contaminated by such low-lying scattering states.

(L4) Existence of the mass gap. This Letter does not address the Wightman cluster-decomposition proof of $m_{\text{gap}} > 0$ in the continuum: we work with the lattice spectrum and derive the dimensionless ratio. The existence layer is addressed elsewhere.

(L5) Lean status. Only K , F_∞ , and ξ_* are Lean-proved; the algebraic identities (3) and (4) are exact rational arithmetic and can be mechanized straightforwardly, but the spectral factor c^2 itself remains empirical at TIER 3 (supported, 14%).

CONCLUSION

We have written down a closed form for the mass-gap ratio m^2/σ_0 of 4D pure $SU(N)$ Yang–Mills in every $J^{PC}(\text{ex})$ channel at every $N \geq 3$, in which every rational coefficient is generated from three geometric anchors ($\xi_* = 2/3$, $F_\infty = 9/10$, $\delta = +2$). The two rationals that the v1 draft left empirical (β and c_η) emerge as exact algebraic identities (3)–(4) in the anchors. The $N \rightarrow \infty$ limit has zero free parameters.

The roadmap is: (i) independent $Sp(2N)$ tests [3] of $\eta(N)$ at the same two derived rationals; (ii) Lean 4 mechanization of (3) and (4) on top of the existing K , F_∞ , ξ_* kernel; (iii) an arithmetic proof of $\delta = +2$ from a flux-tube flexion-vibration partition.

Data availability. Lattice data are from [2, 3]; worksheets are at the project repository. All numerical work was carried out in PARI/GP 2.15 with cross-checks in Python; the Lean 4 proofs of K , F_∞ , ξ_* are kernel-verified at commit `Crossed/Transport.lean`.

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